MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

January 2021

Mathematical Methods

Module Leader: Dr Layal Hakim

Duration: 2 HOURS (+30 minutes upload time)

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN NOTE** examination

SECTION A

1. (a) Find the volume of the parallelepiped in $\mathbb{R}^3$ built on the vectors $(1, -1, 1)$, $(1, 1, 1)$, and $(1, 2, 4)$. (10)

   (b) Use a suitable convergence test to determine if the following series converges

$$\sum_{n=1}^{\infty} \frac{3n^2 + 2n}{2^n}.$$

(10)

   (c) Compute the sum

$$\sum_{k=3}^{\infty} \frac{3}{2^k}.$$

(7)

   (d) Find the values of a and b that make $f(x)$ continuous everywhere, where $f(x)$ is given by

$$f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & \text{if } x < 1 \\ ax^2 + bx, & \text{if } 1 \leq x \leq 2 \\ 2ax + b + 6, & \text{if } x > 2. \end{cases}$$

(10)

   (e) Calculate the limits

   (i)

$$\lim_{x \to \infty} -x + \sqrt{x^2 + 3x}$$

   (ii)

$$\lim_{x \to \infty} \frac{7x^4 + x^2 - 2}{14x^4 + 5x - 4}.$$

(13)

[**50**]

SECTION B

2. (a) Let f be defined by $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \cos(x)$ and $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \sin(x)$.

   (i) Write down the definition of the function h given by the composition $f \circ g$.

   (ii) Express $h(x) = (f \circ g)(x)$ as a Taylor series about $x = 0$ giving terms up to and including x^4. (10)

   (b) Let

$$y = (1 + 2x)e^{-x}.$$

   Sketch the curve of y by finding any stationary points and any points of inflexion, as well as the limiting behaviour of x. Indicate any points where the curve crosses the axes. (15)

[**25**]

3. (a) Find

$$\lim_{x \to 0^+} x^{x^x - 1}.$$

(8)

   (b) Show that the slope of the tangent to the cardioid $r = 1 - \sin(\theta)$ can be written in the form

$$\frac{dy}{dx} = \frac{\sin(2\theta) - \cos(\theta)}{\cos(2\theta) + \sin\theta}.$$

(9)

   (c) Let $p_1x + q_1y + r_1 = 0$ and $p_2x + q_2y + r_2 = 0$ be the equations of two perpendicular lines. Show that $p_1p_2 + q_1q_2 = 0$. (8)

[**25**]

4. (a) Determine the values of x for which the following series converges

$$\sum_{k=0}^{\infty} \left(\frac{\alpha(\alpha-1)...(\alpha-k+1)}{k!} \right) x^k$$

where $\alpha \in \mathbb{R}$ is not an integer. (7)

(b) Give an example of a function that is continuous everywhere on its domain, but fails to be differentiable at $x = 4$ and show that it is not differentiable at this point using differentiation from first principles. (10)

(c) The Cauchy-Schwartz inequality states that if $\vec{u}, \vec{v} \in \mathbb{R}^3$ then $\vec{u} \cdot \vec{v} \leq |\vec{u}||\vec{v}|$ where the equality holds (i.e. $\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}|$) if and only if $\vec{u} = \lambda \vec{v}$ for some real number λ. By applying the Cauchy-Schwartz inequality to vectors $\vec{u} = (x, y, z)$ and $\vec{v} = (y, z, x)$, show that

$$x^2 + y^2 + z^2 \geq xy + yz + zx$$

for all $x, y, z \in \mathbb{R}$. For which x, y, z does the equality hold? Justify your answer. (8)

[**25**]